

REPORT – EXPLAINABLE AI 2026

INTRODUCTION

The domain of this study is anomaly/fraud detection in the domain of credit cards and fraudulent transactions.

In many financial systems, customers will receive the message of an anomaly being detected, but more often than not, said message will not explain why or at which point of the transaction the anomaly was individuated. This makes it difficult for customers or bank/financial institution employees to understand what went wrong, and therefore, how to prevent the same thing happening in future transactions. Delays in understanding the problem will also often lead to a significant decrease in efficiency of fraud investigations ²(Eleanor R. et al., 2023).

Explainable AI can help with explaining to humans what and where the anomalies are happening, and most importantly, why ³(Molnar, 2022). For example, it could state the exact feature that caused the anomaly, and show a scenario in which the transaction happened with no anomalies, in order to find the changes to make for a legitimate transaction.

It is however unsure whether XAI can always be the solution to understanding fraudulent transactions when the elements and features of such transactions are anonymized by functions of, for example, privacy of the customer.

However, to conclude on a positive note regarding AI systems and their implementation: as long as these systems are used in a completely transparent way, humans will start to increasingly trust them, leading to illegitimate transactions being solved more efficiently with no wasted time in between ⁴(Lundberg & Lee, 2017).

RESEARCH QUESTION

The logical research question is:

“While Explainable AI methods are approved tools for identifying and solving anomalies in fraudulent transactions, how actually useful are they for human operators when the features in the data are anonymized and nonactionable?”

Which allows us to experiment with XAI methods, to prove that they can help explain anomalies and suggest future improvements, but most importantly to switch the focus on whether it still makes sense to employ such machines when the anomalous features are anonymous/censored and/or non-actionable (fixed data points that a human cannot change, like the time of a transaction).

The dataset I decided to use is an anomaly dataset ¹(Arshahuja, 2019) that detects fraud happening with Credit Cards. It is large-scale, with over 280k sets of data, which will allow in-depth training for the XAI methods.

However, the only actionable and visible data point is “Amount” (of the transaction). The rest is either nonactionable like “Time” (of the transaction), or anonymous like features V1-28, which we do not know what they represent.

The research question stems exactly from such characteristics of the selected dataset. Many datasets in real-life instances have censored or anonymous features, which raises the question to whether, at the end of the day, ex-AI methods are still useful in scenarios where operators need to learn from errors and know how to properly operate in the future.

METHODS

To try and answer our question, we will first analyze the dataset and evaluate each feature’s contribution to fraudulence through the XAI method of SHAP. We then proceed by creating scenarios in which those features do not cause to detect any fraudulence through the counterfactual explanations XAI method. Here is a short description of the two methods:

SHAP (Shapley Additive exPlanations): this is a method based on cooperative game theory that “awards points” to each feature, indicating the weight of the feature’s impact to the model’s final prediction of fraudulence. The rewards are, more formally, Shapley values that mathematically represent the average contribution of each feature across all feature combinations. SHAP relies on accuracy and consistency, and to ensure correct Shapley values estimation for the Random Forest model, using the TreeExplainer resulted in a better architecture choice.

Counterfactual Explanations (DiCE): this is a method that shows which changes in input features would allow the transaction to not be fraudulent anymore. It operates in a what-if manner, finding the smallest possible feature adjustments in order to turn any anomalous transaction (with “Class” feature of value 1), into a non-anomalous one (“Class” value of 0). It is after these adjustments that feasibility from a human operator is evaluated.

Ultimately, the two methods operate for different purposes: the first to evaluate the weight of each feature to fraudulence detection, with its strength being the mathematical accuracy of features values;

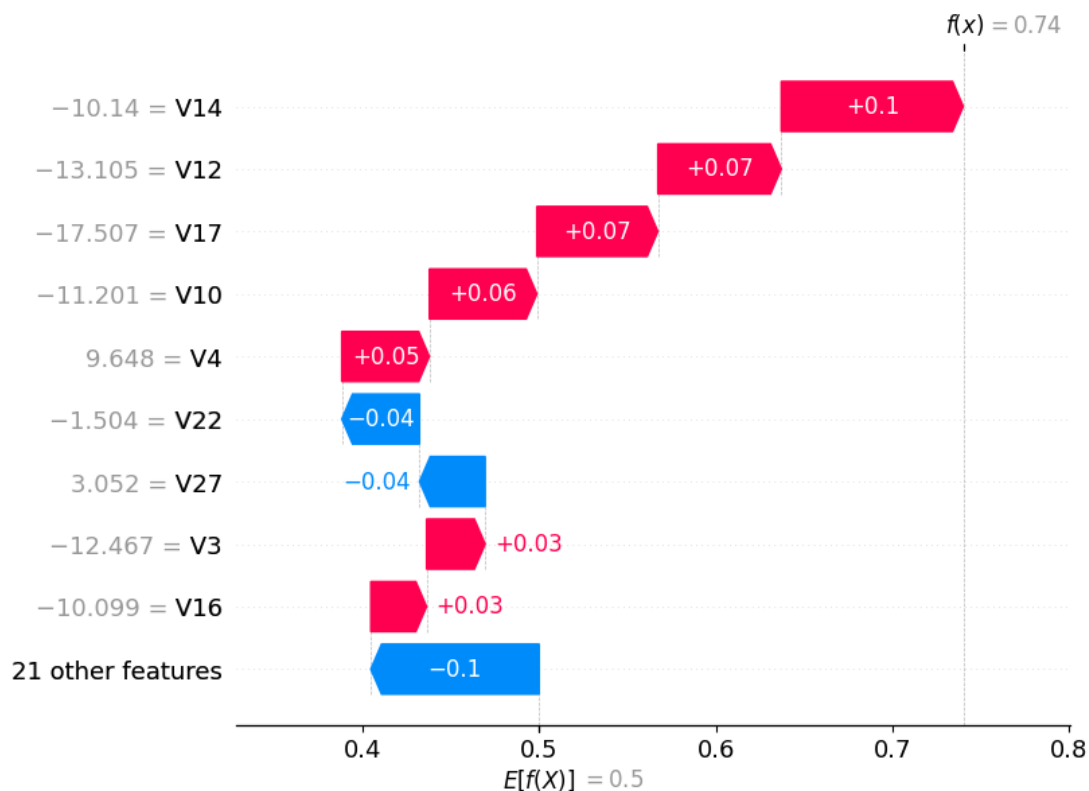
the second to evaluate the ideal value of the (ex-)fraudulent feature, with its strength being multiple accurate scenarios for the user’s interest. The weakness of both the methods is that both work assuming that such features are nominative and recognizable by a human operator when, in factuality, they are censored most of the time. And so, regardless of their accuracy in performance, they lack usefulness.

RESULTS

Before applying the explainable-AI methods, I first trained a Random Forest Classifier on the chosen dataset `creditcard.csv`. Which achieved a near perfect accuracy of 0.9995 on the test set. From here, for the purpose of the methods, it was better to isolate only the anomalous transactions (of Class = 1).

SHAP results

Using `shap.TreeExplainer`, it was then possible to generate the waterfall plot of the first (because of `frauds.iloc[[0]]`) fraudulent raw transaction.



It seems that SHAP performed perfectly, with high accuracy in identifying the anomalous-heavy features. The final score is $f(x) = 0.74$, meaning a 74% confidence in the evaluation. In this instance, then most relevant features are V14 (+0.1) and V12 (+0.07).

This shows a strong mathematical accuracy and importance, especially in reaching the final value. However, as mentioned before, there is an evident pitfall: how is a human operator supposed to understand how to change the feature values if these values are anonymized like in most bank systems? This makes the entire function of SHAP evaluations ultimately useless.

Counterfactual Explanations results

If SHAP is useless in being able to understand the necessary changes to make a fraudulent transaction not one, counterfactual explanations does exactly that. With the `dice_ml`⁵(Ramaravind K. Et al., 2020) library we generated 3 “what-if” scenarios.

Query instance (original outcome : 1)

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22	V23	V24	V25	V26	V27	V28	Amount	Class
0	18088.0	-12.224021	3.85415	-12.466765	9.648311	-2.726961	-4.445611	-21.922812	0.320792	-4.433162	...	-1.15983	-1.504119	-19.254328	0.544867	-4.781606	-0.007772	3.052358	-0.775036	1218.890015	1

1 rows × 31 columns

Diverse Counterfactual set (new outcome: 0)

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22	V23	V24	V25	V26	V27	V28	Amount	Class
0	-	-	-	-	-	-	-	-	-	-	...	-	-	-	-	6.4860838	-	-	-	-	0.0
1	-	-	-	-	-	-	-	-	-	-	...	-	-	-	-	-	-	-	-	-6.559863472	0.0
2	-	-	-55.3032679	-	-	-	-	22.73602561	-	-	...	-	-	-	-	-	-	-	-	-	0.0

3 rows × 31 columns

As it shows, out of the original row with outcome “Class” = 1 (fraudulent), three diverse Counterfactual sets are shown with outcome “Class” = 0 (non-fraudulent). In the first instance, a change in the value V25 solves the problem. The changes can also be more than one, like in the third instance with V2 and V7. What is important is that “Class” turns into a 0. However, the problem is, and if the pattern was already picked up might be obvious, the actual meaning of such scenarios. A human operator will not be able to do two things:

- 1- understand the reason behind such changes for future reference;
- 2- actually change the features’ values as some of them might be a customer’s private unchangeable info, or possibly unchangeable measures like the amount of the transaction.

DISCUSSION

Pitfalls and biases

The results of both SHAP and DiCE, despite accurate and faithful to what they were created for, they show a fundamental pitfall in the financial world: the illusion of usefulness. An operator can surely be equipped with explainable AI models and tools, but it does not mean that the results will be interpretable and, therefore, useful for future instances of the same problems occurring. These tools are transparent in the technicalities and operations, but they are completely blurred in the practice whenever data is anonymous or obscured.

At the same time, if bank and financial operators are provided with such tools, it is easy to assume the potential growth of Automation Bias, where automation and its output is trusted too much. This may lead to blindly flagging fraudulent-detected transactions, without real understanding of the reason, and therefore without the possibility of the usually granted dispute from the flagged customer(s).

Ethical considerations

When customers are denied protest, a serious ethical issue takes place. Customers have the fundamental right to explain any assumption made in their regard, even if such assumption was made from an automated system. On the other hand, solving this problem would mean opening the privacy restrictions that privatize features, which, obviously, would have impact on the privacy of the customers since now all their personal data has been entrusted into a machine with no real understanding of privacy.

CONCLUSION

In conclusion, the Explainable AI tools seem to not be compatible with this human-driven world of finance. On one side, they are only useful in solving the problem, while failing at explaining it, which leads to improper professional conduct by operators, and lack of customers' rights. On the other hand, to solve the explainability problem, it would mean to violate a different customer's right, privacy, by removing the anonymization of the features. In the end, to be able to see Explainable AI tools in the finance world, new frameworks should be developed, where automatization will only make mistakes clearer to operators, while maintaining all customers' rights intact.

REFERENCES

- ¹ Arshahuja, Anomaly Detection, *Kaggle*, (last updated: 2019).
<https://www.kaggle.com/datasets/arshahuja/anomaly-detection>
- ² Eleanor R. et al, (2023). "Opportunities in Real Time Fraud Detection: An Explainable Artificial Intelligence (XAI). Research Agenda", *International Journal of Advanced Computer Science and Applications*, Vol.14(5), pp.1172- 1186.
<https://doi.org/10.14569/IJACSA.2023.01405121>
- ³ Molnar, C. (2022). *Interpretable machine learning* (2nd ed.), Lulu.com.
<https://christophm.github.io/interpretable-ml-book/>
- ⁴ Lundberg S. M., & Lee S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
<https://doi.org/10.48550/arXiv.1705.07874>
- ⁵ Ramaravind K., Mothilal, Amit S., Chenhao T., (2020). Diverse Counterfactual Explanations (DiCE) for ML. <https://interpret.ml/DiCE/>

CODE LINK

https://github.com/matildeliotti/Explainable_ai_final/tree/main